

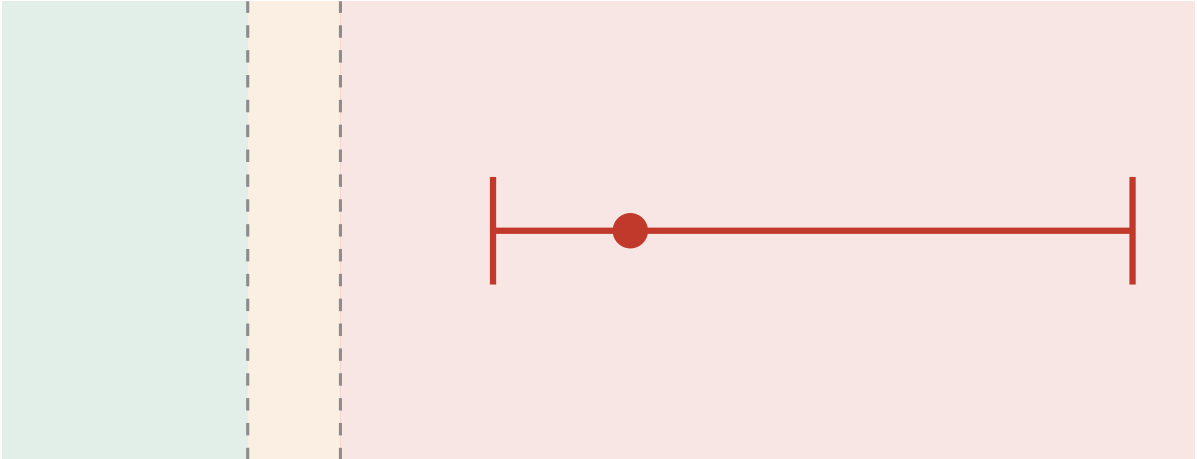
# Bias-effect ratio gauge

BER = 8.73, 95% CI [3.13, 372.94] -> bias-dominated

competitive

effect-dominated

bias-dominated



BER = 8.73

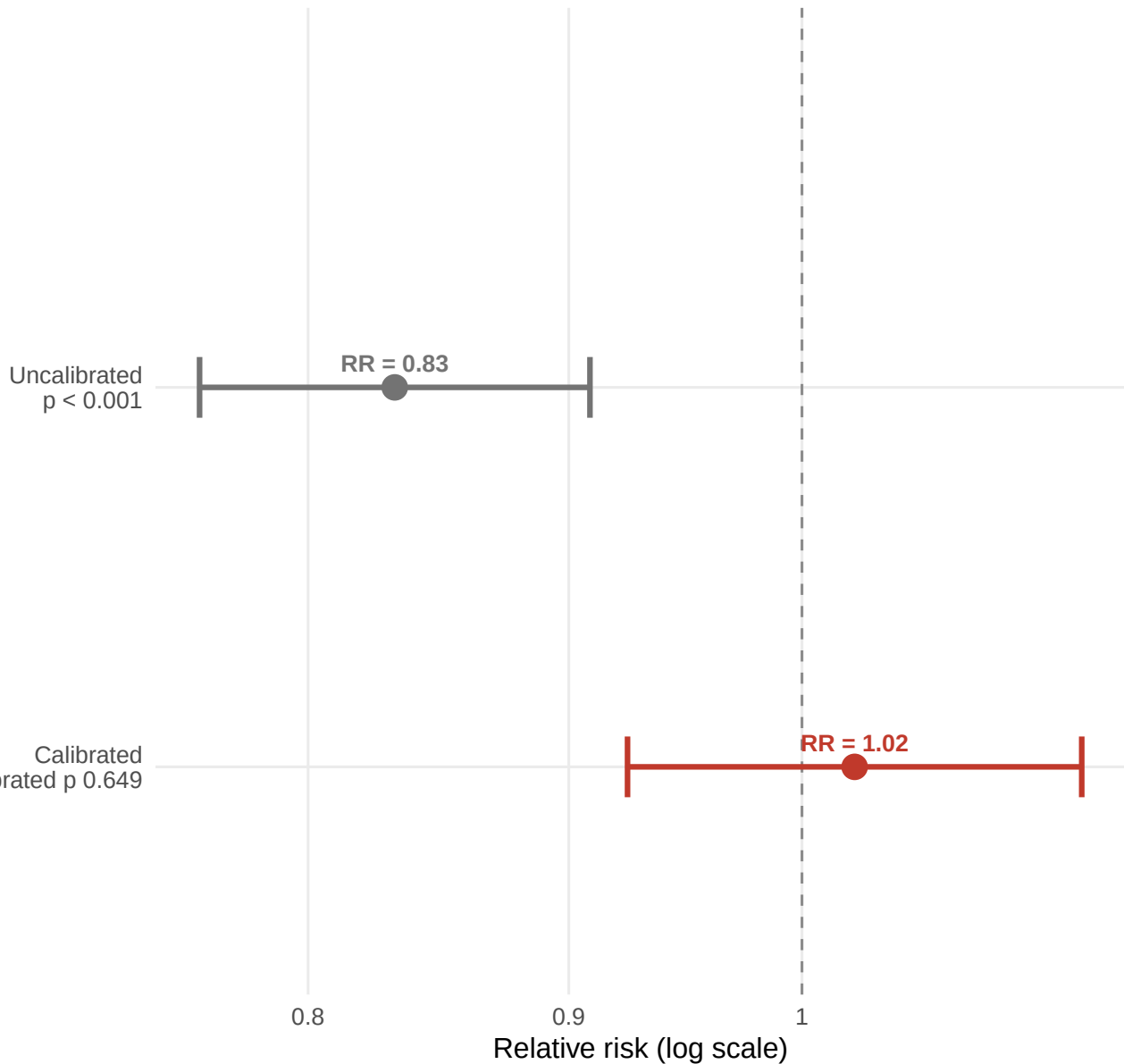
0.1                      0.5                      1                      2                      10                      100

BER (log scale)

Zones: BER < 0.5 effect-dominated | 0.5–1 competitive | > 1 bias-dominated

# Effect estimate before vs after empirical calibration

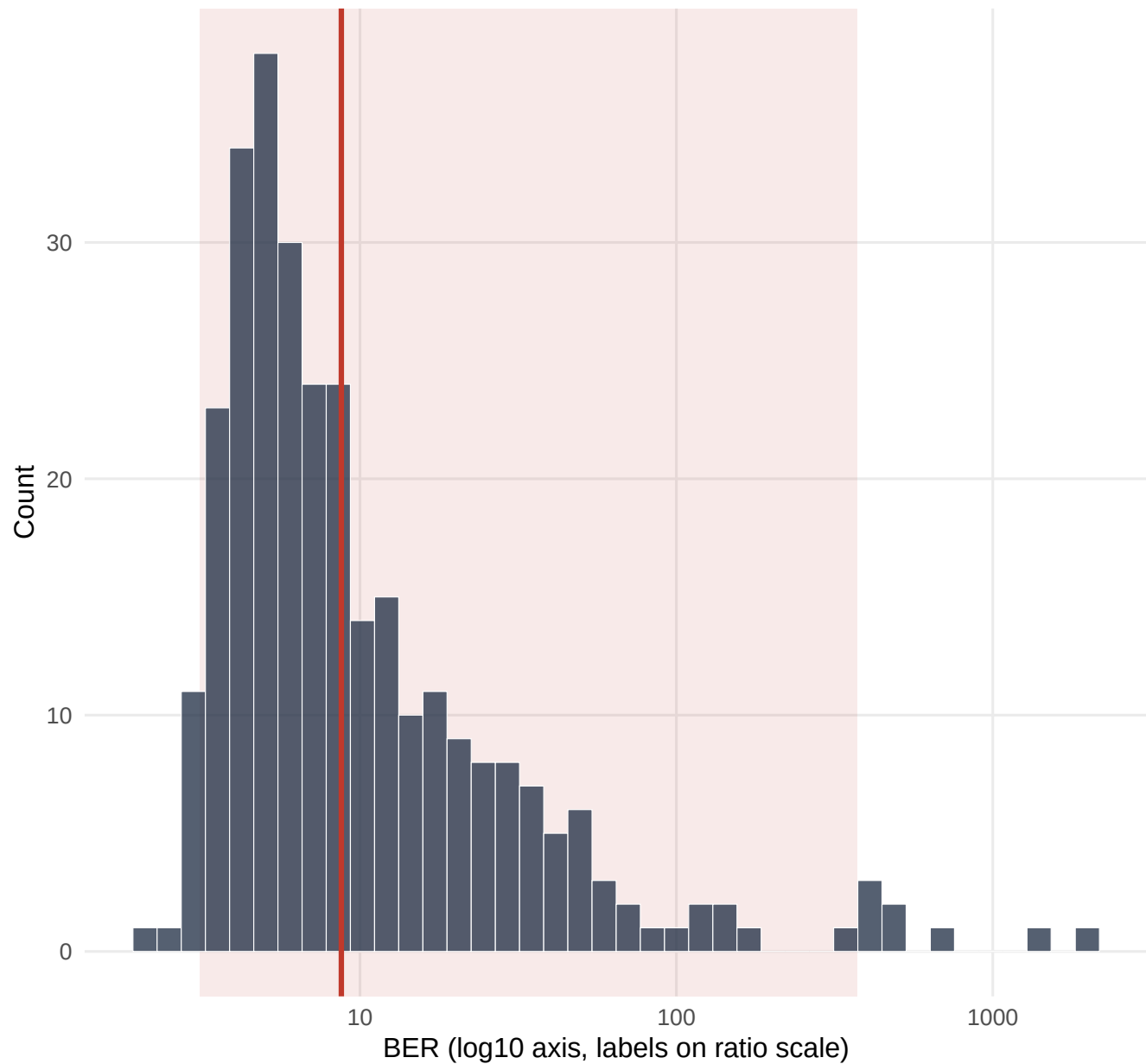
BER = 8.73  $\rightarrow$  bias-dominated; calibrated CI incorporates systematic error ( $\sigma_E$ )



Dashed line: no effect (RR = 1)

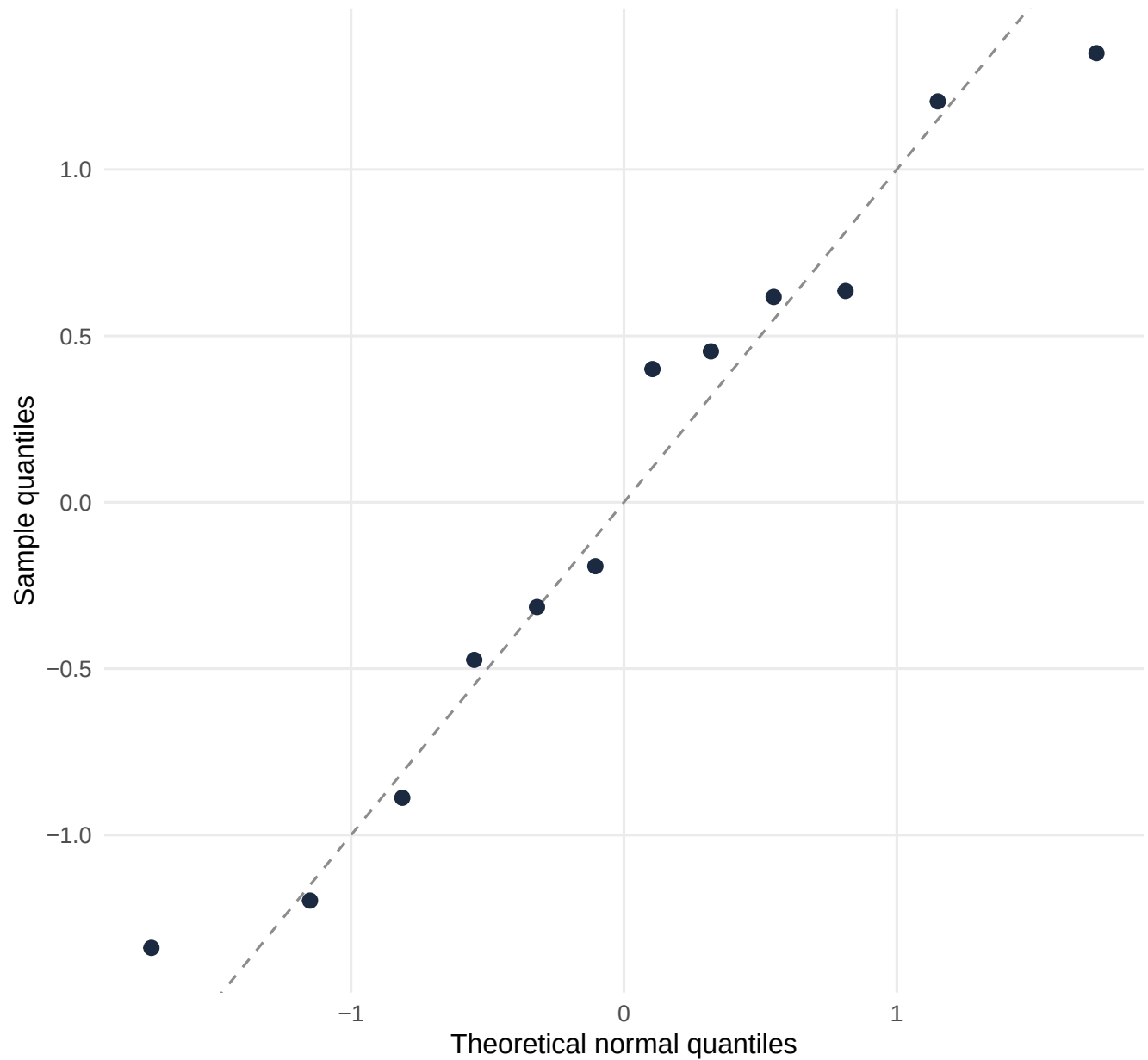
# Bootstrap distribution of the BER

300 successful replicates; shaded band = 95% CI [3.13, 372.94]; line = point estimate



# Q-Q plot of standardized negative control residuals

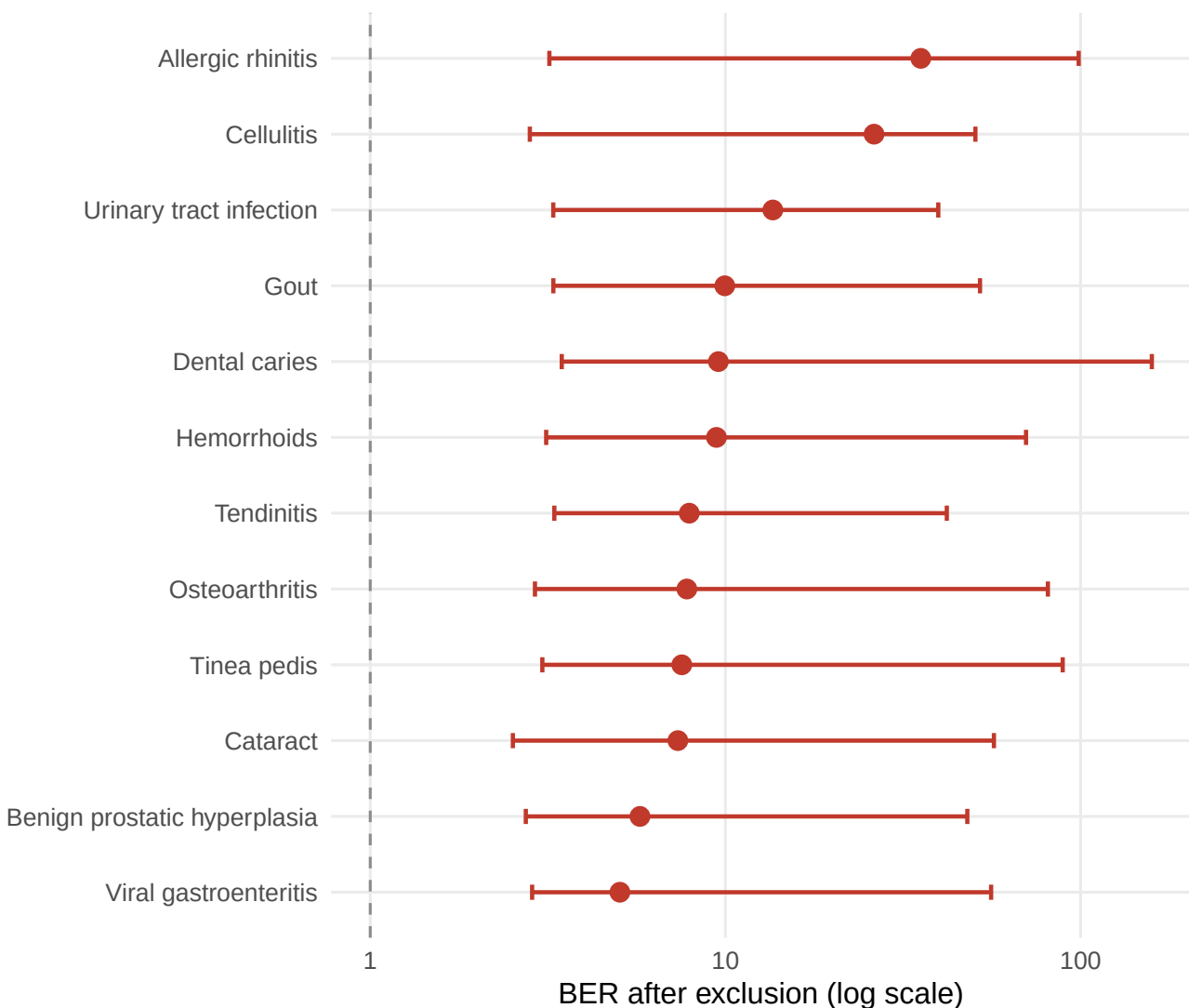
Shapiro-Wilk p = 0.637; residuals standardized by marginal SD  $\sqrt{\text{se}^2 + \sigma_B^2}$



# Leave-one-out sensitivity of the BER

Each row excludes one negative control and refits the empirical null

Excluded negative control



—●— bias-dominated

Dashed line: BER = 1 (bias equals calibrated signal)